Detecting Synthetic Facial Images

Machine Learning 1 - Final Presentation

1TRUTH

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Table of Contents

- Problem Definition and Project Overview
- Data Processing
- Modeling Methodology
- Key Findings
- Conclusion

Problem Definition and Project Overview

Problem Definition

Can machine learning detect synthetic facial images using learned visual patterns?

- Advances in generative AI allow highly realistic synthetic facial images to be created from simple prompts
- Synthetic Facial images are increasingly used in fraud, impersonation, fake profiles, and misinformation
- Peer-to-peer platforms (social media, dating apps, online profiles), often lack verification tools
- It is becoming increasingly difficult to distinguish real human faces from AI-generated images

Project Overview

Goal: Develop a machine learning classification model to detect synthetic facial images

- Dataset composed of real and AI-generated facial images
- Extracted face embeddings to represent each image as numerical features.
- Applied unsupervised analysis to explore feature structure.
- Built supervised classification models to distinguish real vs synthetic faces.
- Selected the best-performing model based on evaluation metrics.
- Applied hyperparameter tuning to optimize the final model.
- Generated authenticity predictions for facial images.

Data Sources

- Fake images: 5,000 samples from the Real vs AI Generated Faces dataset.
- Real images: 5,000 samples from the Flickr-Faces-HQ (FFHQ) dataset.
- Fake image categories: Faceswap, SFHQ (Synthetic Faces High Quality), Stable Diffusion, and ThisPersonDoesNotExist.

Faceswap

SFHQ

Stable Diffusion

ThisPersonDoesNotExist

Data Processing

Data Preprocessing

Image Embeddings

- Convert high-dimensional images into **512 features** while preserving semantic information (compact, informative vectors)
- Similar human faces (e.g., skin tone, hair color, facial structure) have closer embeddings
- Embeddings can be used for:
    - Classifying real vs. fake images
    - Clustering similar faces

Implementation:

- ResNet18 pre-trained on ImageNet
- 512-dimensional embeddings fed into a ML models for image classification
- Label: **0 for fake and 1 for real.**

emb_0	emb_1	emb_2	...	emb_510	emb_511	Label
0	0	1.055535		1.144806	0.955699	0

Dimension Reduction

Approach:

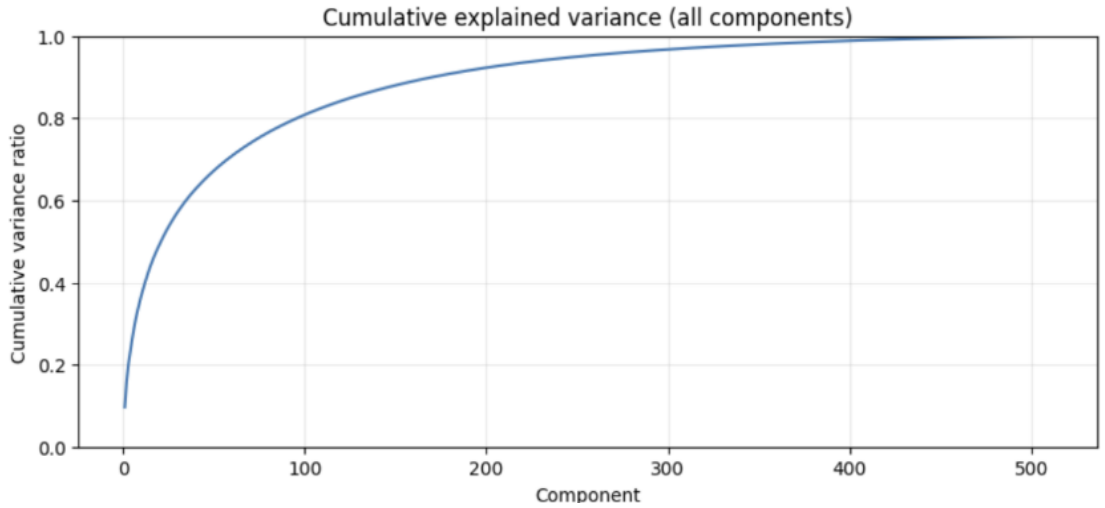
- Applied Truncated Singular Value Decomposition (TruncatedSVD) to identify orthogonal components capturing the largest variance.

Key Findings:

- ~21 components → **50% variance captured** (compact representation).
- ~96 components → **80% variance captured** (higher-dimensional, more information preserved).
- The cumulative variance curve rises quickly, showing **most variance captured by few components**.

Selected Representations for Experiments:

- **Variance > 50 %:** SVD20
- **Variance > 80 %:** SVD100
- **Variance > 90 %**: SVD200
- Full Dimension (512 features)

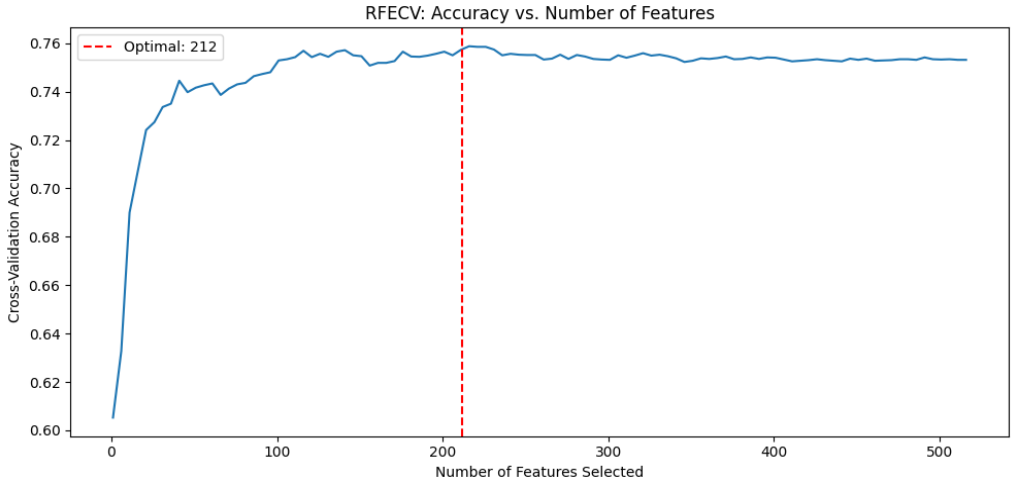


Reduced Feature Set Exploration

Goal: Improve Efficiency Without Losing Accuracy

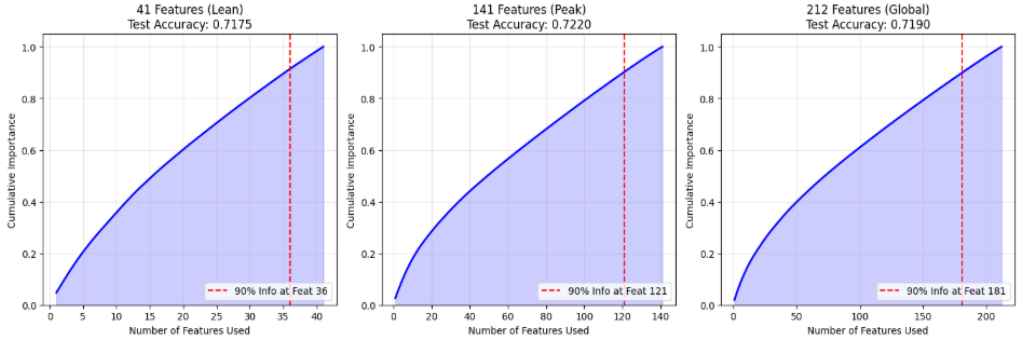
Method: RFECV (Recursive Feature Elimination with Cross-Validation)

- Starts with **all 512 features** and ranks them by importance.
- Iteratively **removes the least important features**.
- Uses **Logistic Regression with cross-validation** to evaluate model accuracy at each step.

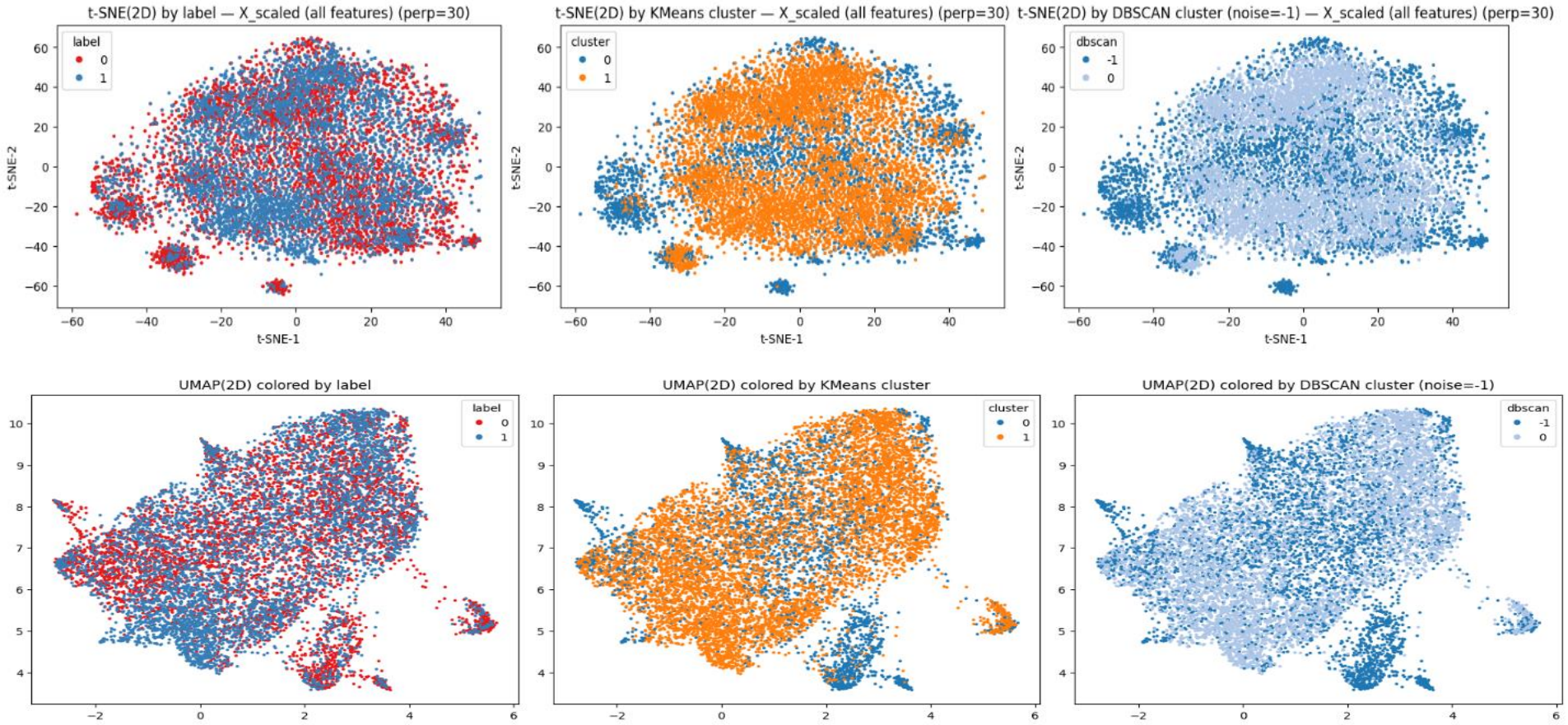


Key Findings

- **Optimal feature count: 212 features** → highest predictive performance.
- **Performance milestones:** accuracy spikes around **41** and **141** features.



Clustering Analysis & Visualization



- **K - Means:** Accuracy of 0.6236
- **DBSCAN:** Accuracy of 0.6504

Modeling Methodology

Candidate Models

Category	Models	Description
Linear Models	Logistic Regression, Support Vector Machine, Linear SVC, Stochastic Gradient Descent, Perceptron	Linear decision boundary. Efficient for high-dimensional features like embeddings.
Tree-Based Models	Decision Tree	Rule-based model that splits data using feature thresholds.
Ensemble Tree Models	Random Forest, Gradient Boosting	Combine multiple decision trees to improve accuracy and reduce overfitting.
Distance-Based Models	K-Nearest Neighbors	Classifies samples based on similarity to nearby data points.
Probabilistic Models	Naive Bayes	Uses probability distributions and feature independence assumptions.
Neural Networks / Deep Learning	CNN, MLP	Learn complex nonlinear patterns. CNN is designed for images.

Model Tuning

XGBoost Hyperparameter Tuning

Trained on **~70K embeddings** (~55/45 imbalance)

Validation on **200 PCA reduced dimensions**

Learning rate: **0.005**, max depth: **6**, mild L1/L2 regularization

Threshold **0.50** → **0.61**; boosts Fake TPR 72% → 77%

SVM Threshold Strategies

Balanced (0.46): TPR: Fake - 0.79, Real - 0.73

Security (0.67): TPR: Fake - 0.92, Real - 0.60

Both yield **76% accuracy** (±0.13)

141-features matched embedding dim performance

Weighted Ensemble Design

SVM + AdaBoost + XGBoost on 141-feature space, blended via validation-based probability weighting.

Model	Val. Accuracy	Weight
SVM (RBF Kernel)	0.78	0.344
XGBoost	0.77	0.337
AdaBoost	0.72	0.318

Model Results Comparison

Model Configuration	Threshold	Accuracy	Fake Prec.	Fake Recall	Real Prec.	Real Recall	Avg F1
XGBoost (Default)	0.50	0.81	0.82	0.72	0.81	0.88	0.81
XGBoost (Tuned Thresh.)	0.61	0.80	0.76	0.77	0.83	0.82	0.80
SVM 141-feat. Balanced	0.46	0.76	0.75	0.79	0.78	0.73	0.76
SVM 141-feat. Security	0.67	0.76	0.69	**0.92**	0.88	0.60	0.75
SVM 512-feat. Security	0.67	0.76	0.70	**0.92**	0.88	0.60	0.75
Weighted Ensemble	0.565	0.77	0.72	**0.87**	0.84	0.66	0.76

Highest Accuracy

XGBoost at default threshold — 80%

Best Fake Detection

SVM Security achieves 92% fake recall

Best Overall Balance

Ensemble: 87% fake recall at 77% accuracy (cost-sensitive)

Core Trade-off

↑ Fake recall ≈ ↓ Real recall by comparable margin across all models

Feature Efficiency

141 features ≈ 512 features — 72% dimension reduction, no performance loss

Ensemble Advantage

Blends complementary learners for flexible error control via threshold tuning

Key Findings

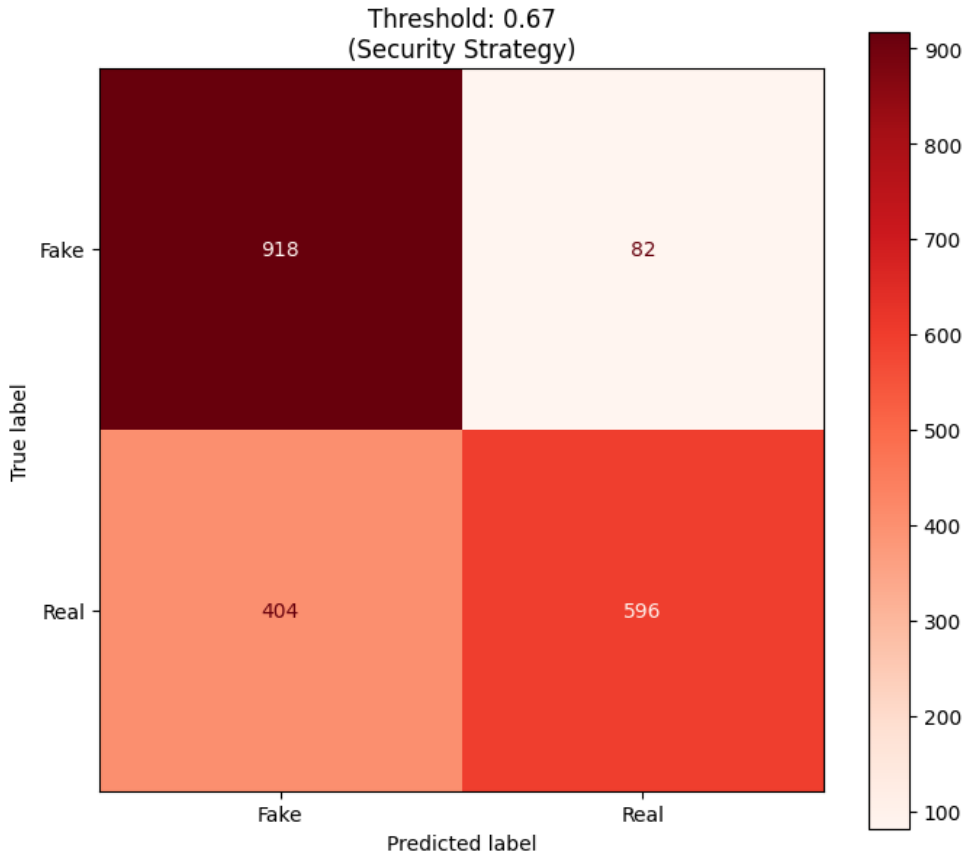
Key Findings

- Learned visual representation can identify synthetic facial images images in peer-to-peer contexts
  - SVM classifies synthetic facial images 92% in a test of 2,000 total images
  - Feature reduction and model tuning can improve model performance
  - Data can be compressed 72% without sacrificing performance
- Embedding data captures interesting patterns that can be used to train smarter models
  - Hats, sunglasses, and microphones are expressed noticeably differently
  - Analysis of embeddings indicates certain features relate to image characteristics

SVM Security Strategy, Social Media Context

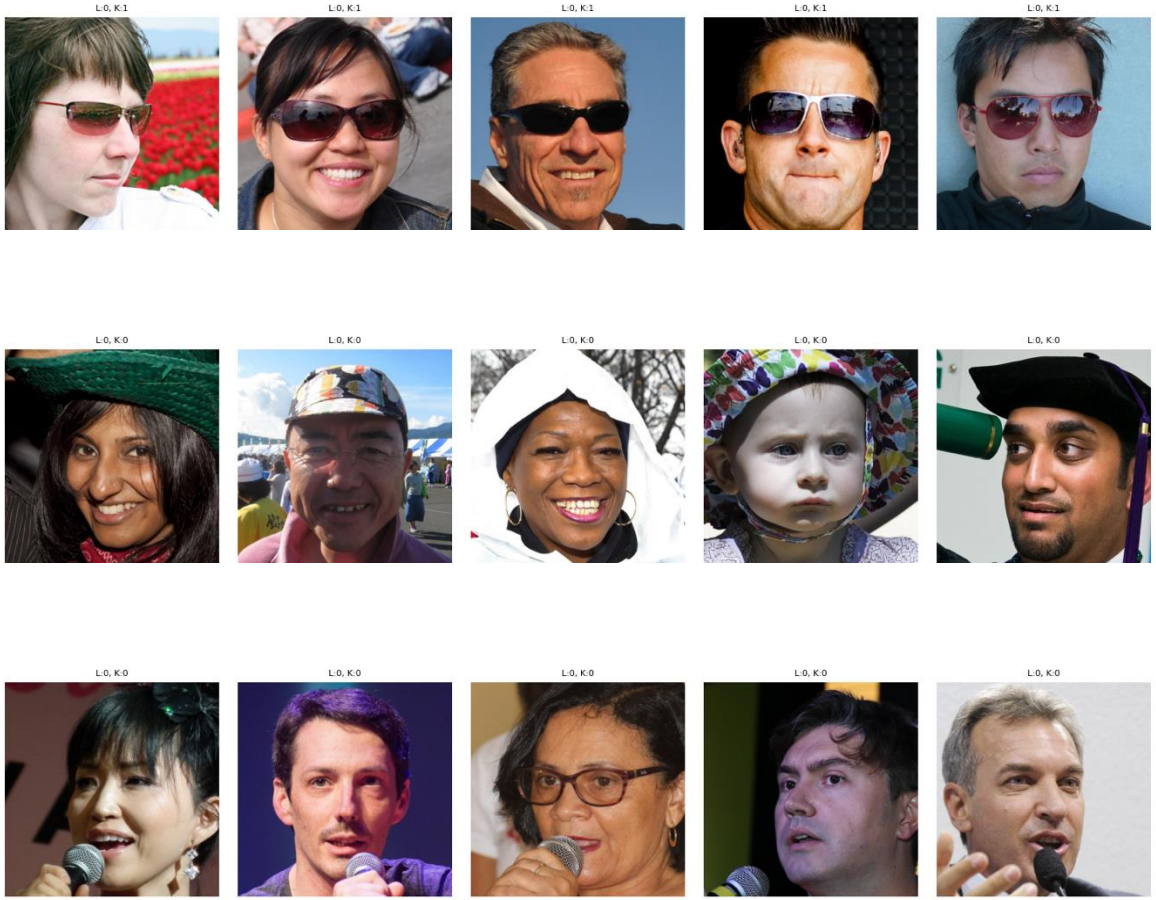
Goal: Develop a machine learning classification model to detect synthetic facial images

- Maximize the correct identification of the Fake class
- Threshold of 0.67: our model must be at least 67% confident of its prediction before assigning a label
  - Correctly classifies Fakes ~92% of the time
  - Misclassifies Real images ~40% of the time
- Security threshold reflects platform logic: suspicious content should trigger caution before trust



Latent Feature Clustering in Image Embedding Space

- The embedding space groups images that share recurring visual characteristics
  - Learned features capture more than overall facial identity
- Distinct accessory patterns emerge across clusters indicating sensitivity to localized visual structure
  - The patterns suggest the model detects interpretable visual signals within high-dimensional feature space rather than relying only on global image similarity
- Future work may identify whether similar clustering exists for facial regions such as eyes, mouth shape, skin texture, or boundary artifacts
  - Its possible to under which specific image characteristics drive synthetic-image detection performance



Next Steps for Future Research

1. Expand and strengthen the training dataset

Incorporate additional authentic facial image datasets to improve model generalization

2. Develop deeper feature-level interpretation

Understand which embeddings correspond to facial regions rather than contextual artifacts

3. Investigate the influence of contextual signals

Ensure the model focuses on intrinsic facial features rather than non-facial patterns

4. Further improve model performance through systematic tuning

Hyperparameter optimization, architecture refinement, and evaluation of deeper models

Overall, this is an area of research that will have continued relevance.

As the line between what is real and generated becomes increasingly blurred, the methods to combat such issues must continue to evolve.

The sophistication of bad actors will continue to increase, and detection systems must follow.

Conclusion

Conclusion

- Lightweight machine learning can reliably detect synthetic facial images, showing that authenticity screening does not require highly specialized systems.
- Facial embeddings preserve enough visual structure to support scalable authenticity assessment in everyday digital platforms.
- As synthetic faces improve, detection must evolve in parallel to preserve trust in digital identity.

"Your face is something you wear all your life, and it tells a very unique story about you…"

Thank You

Questions?

Executive Summary

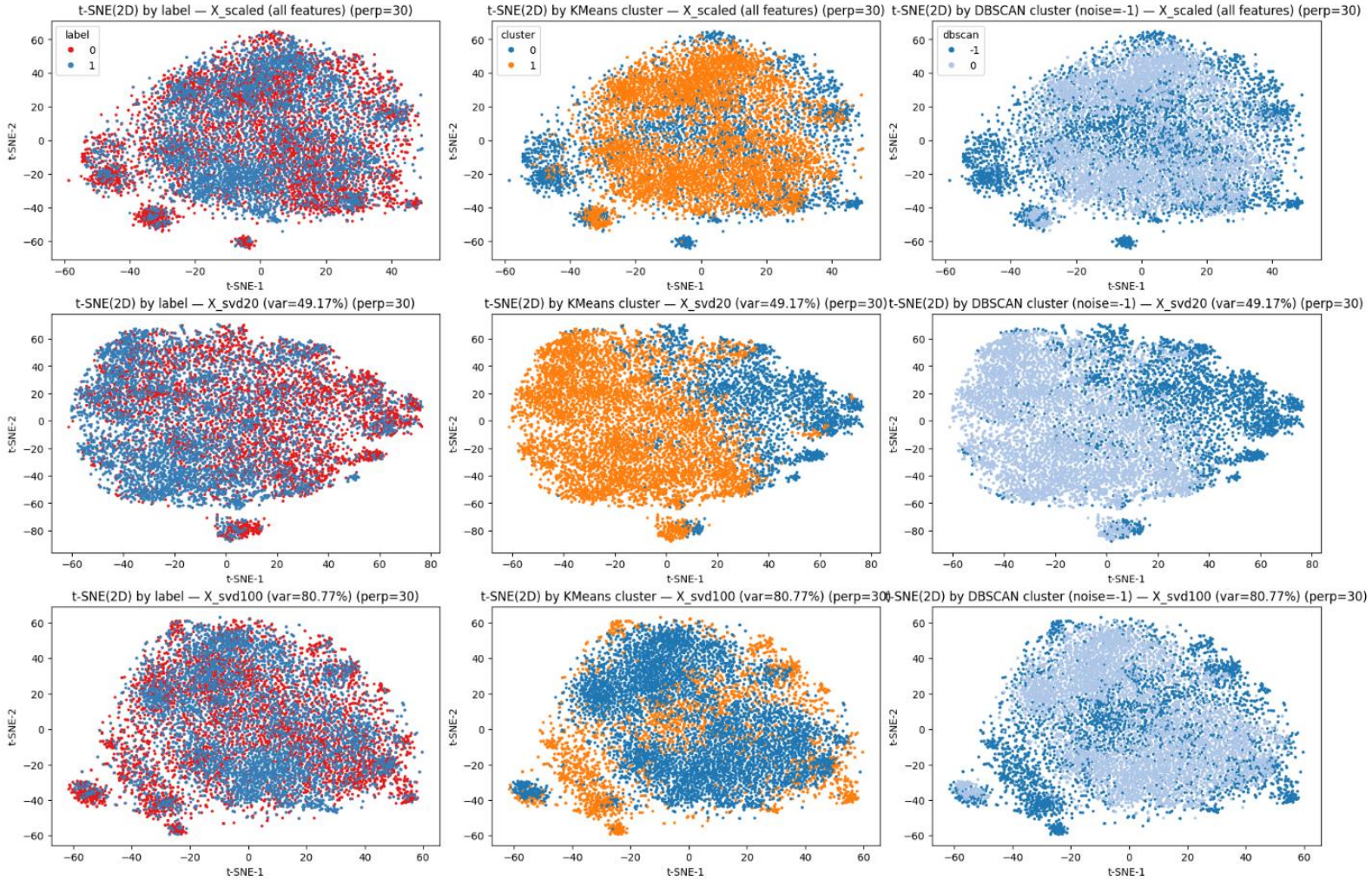
- Learned visual representations can distinguish real vs AI-generated facial images: Face embeddings contain sufficient information to detect synthetic faces.
- Unsupervised analysis revealed contextual patterns: Certain objects (hats, glasses and microphones) cluster together, indicating models may capture contextual signals in images.
- Supervised models achieved strong performance: Models such as SVM and XGBoost achieved over 75% accuracy in detecting synthetic faces.
- Feature reduction improved efficiency: Reducing the embedding space from 512 to 141 features preserved performance while reducing data by ~72%.
- Lightweight models are practical for real-world deployment: Traditional models performed competitively with neural networks and are suitable for scalable authenticity screening tools.
- Machine learning provides a scalable approach for supporting trust in facial imagery within modern digital communication environments.

Appendix

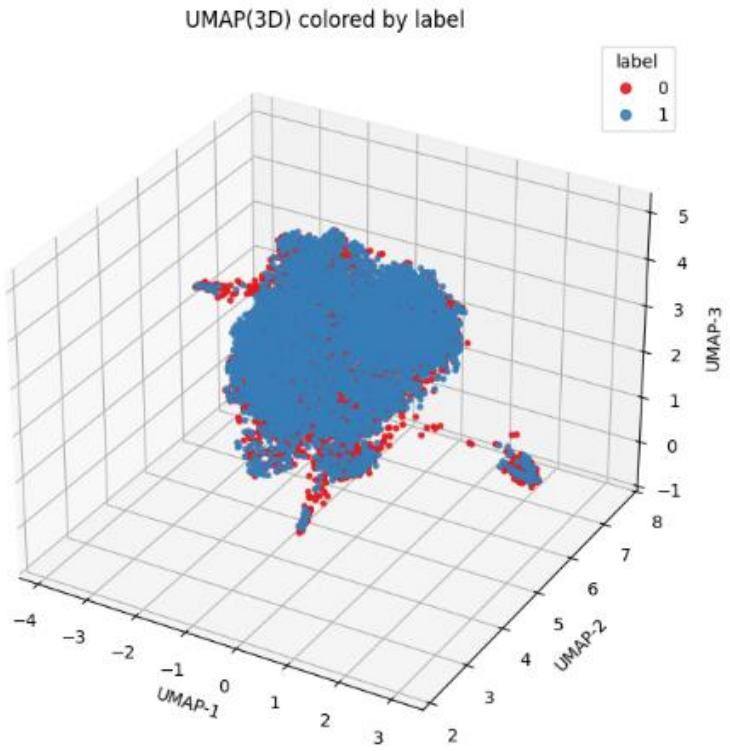
Sources

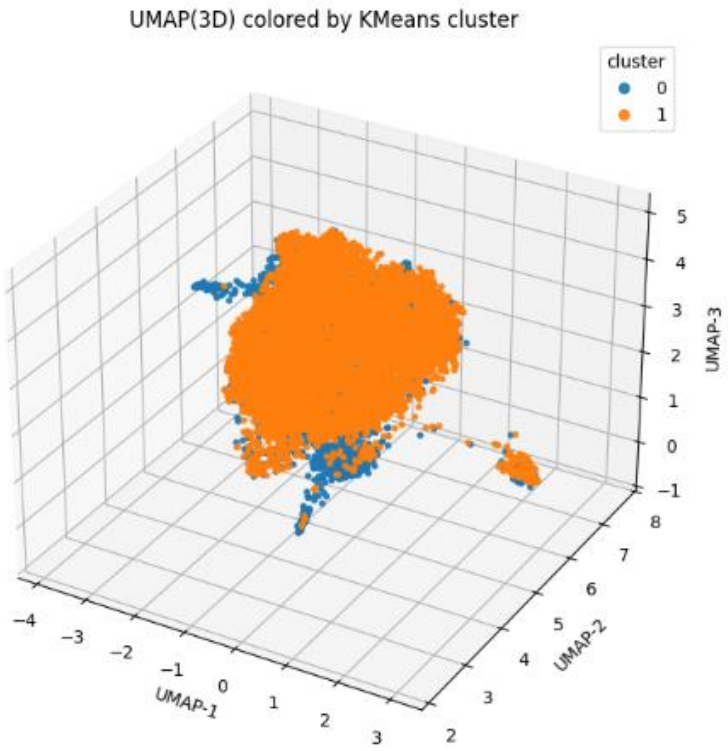
- Swipe left on fraud risk - Detecting and deterring romance scams
  - Moody's
- Deepfake AI Detection Market Size and Competitive Analysis by 2031
  - The Insight Partner's
- On the Detection of Digital Face Manipulation
  - Hao Dang, Feng Liu, Joel Stehouwer, Xiaoming Liu, and Anil Jain
  - Department of Computer Science and Engineering, Michigan State University
- Seeing is no longer believing: Artificial Intelligence's impact on photojournalism
  - David Carson (JSK Journalism Fellowships at Stanford)
- A Style-Based Generator Architecture for Generative Adversarial Networks
  - Tero Karras, Samuli Laine, Timo Aila (NVIDIA)
- Real vs AI Generated Faces dataset
  - MIT, Kaggle

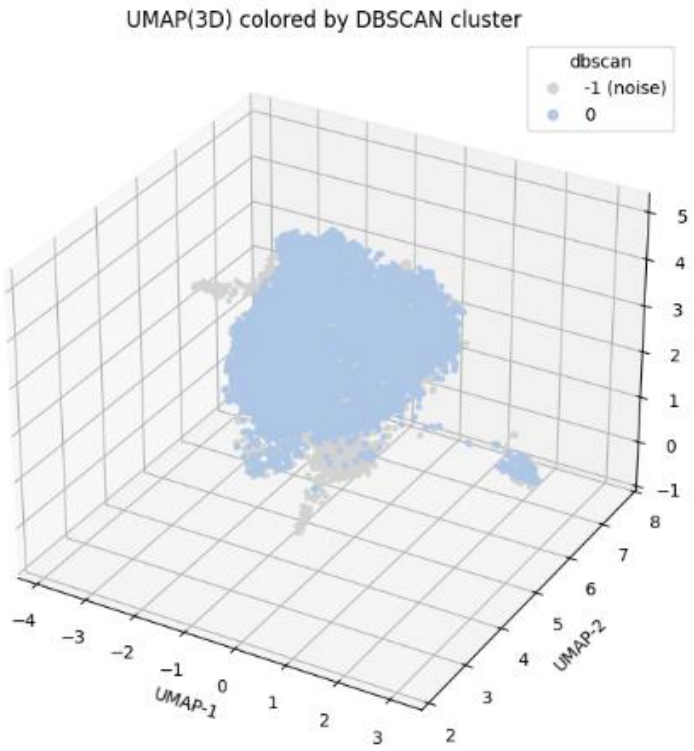
Item 1 - Clustering Visualizations (t-SNE)



Item 2 - Clustering Visualizations (3D UMAP)

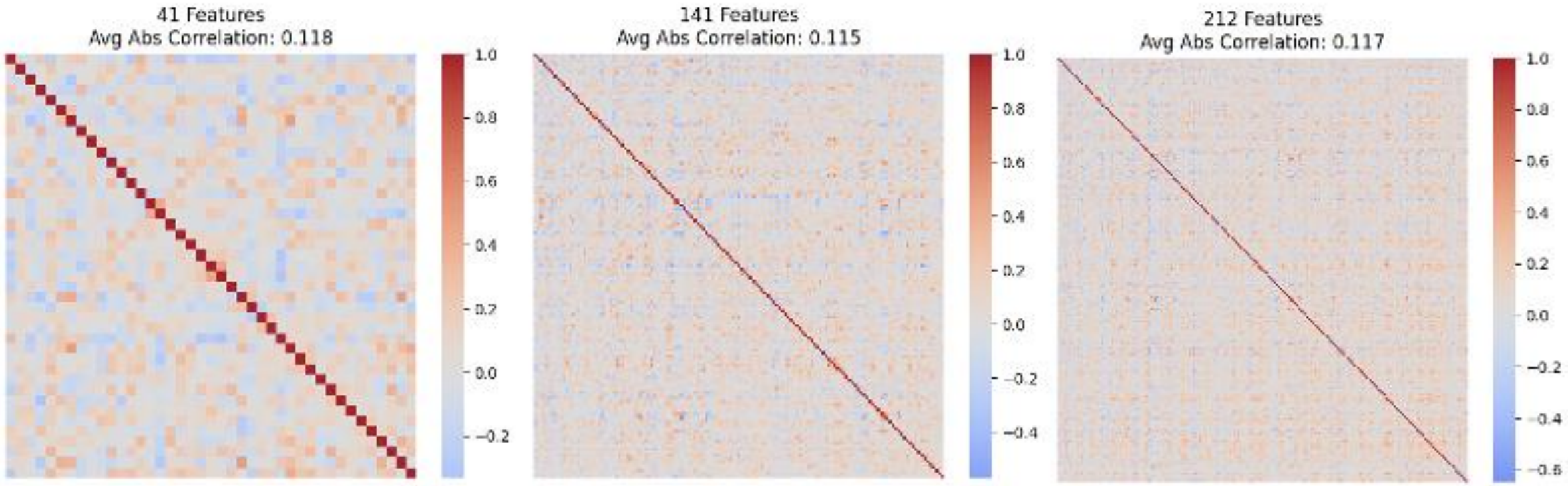




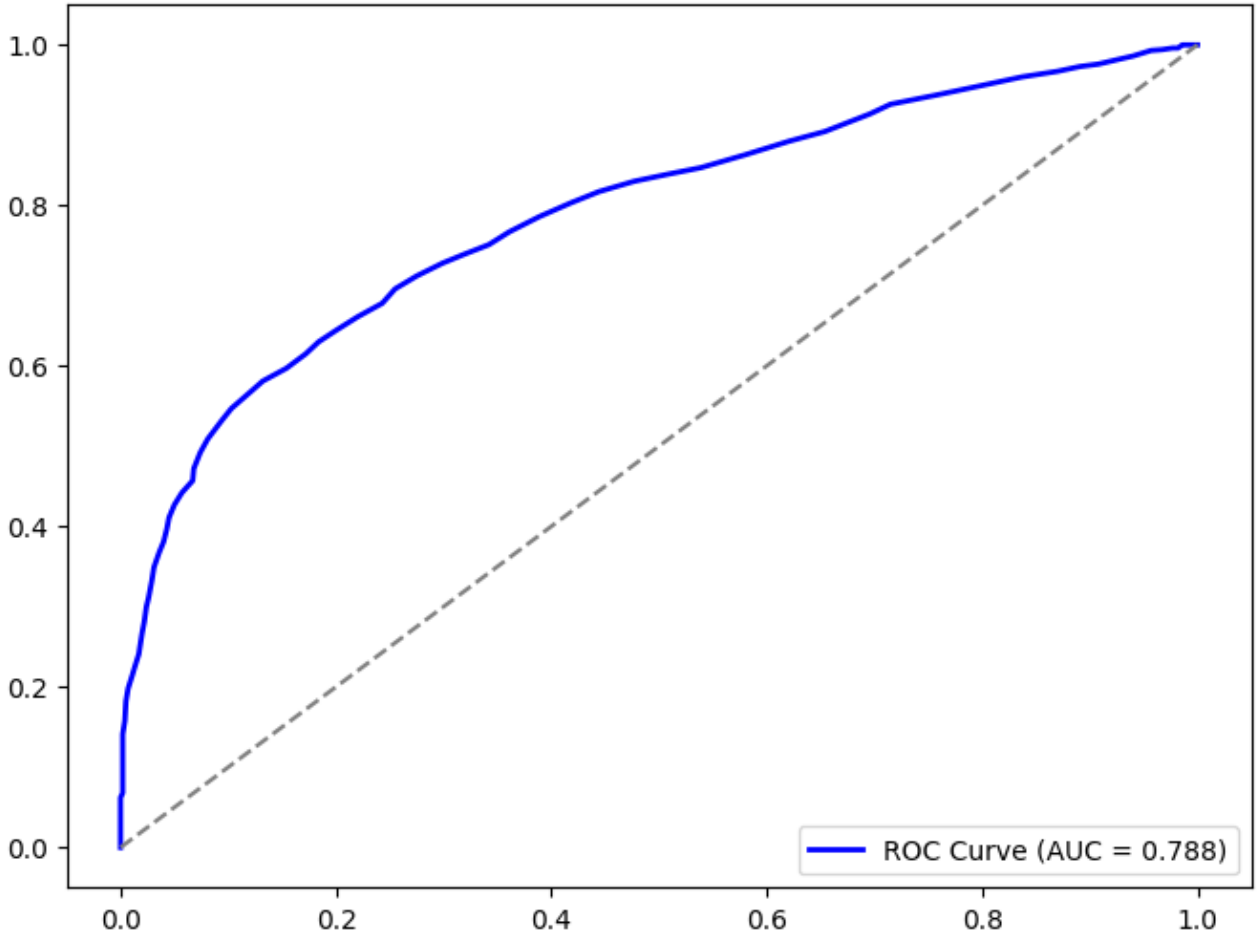


Item 3 – Expression of Specific Characteristics

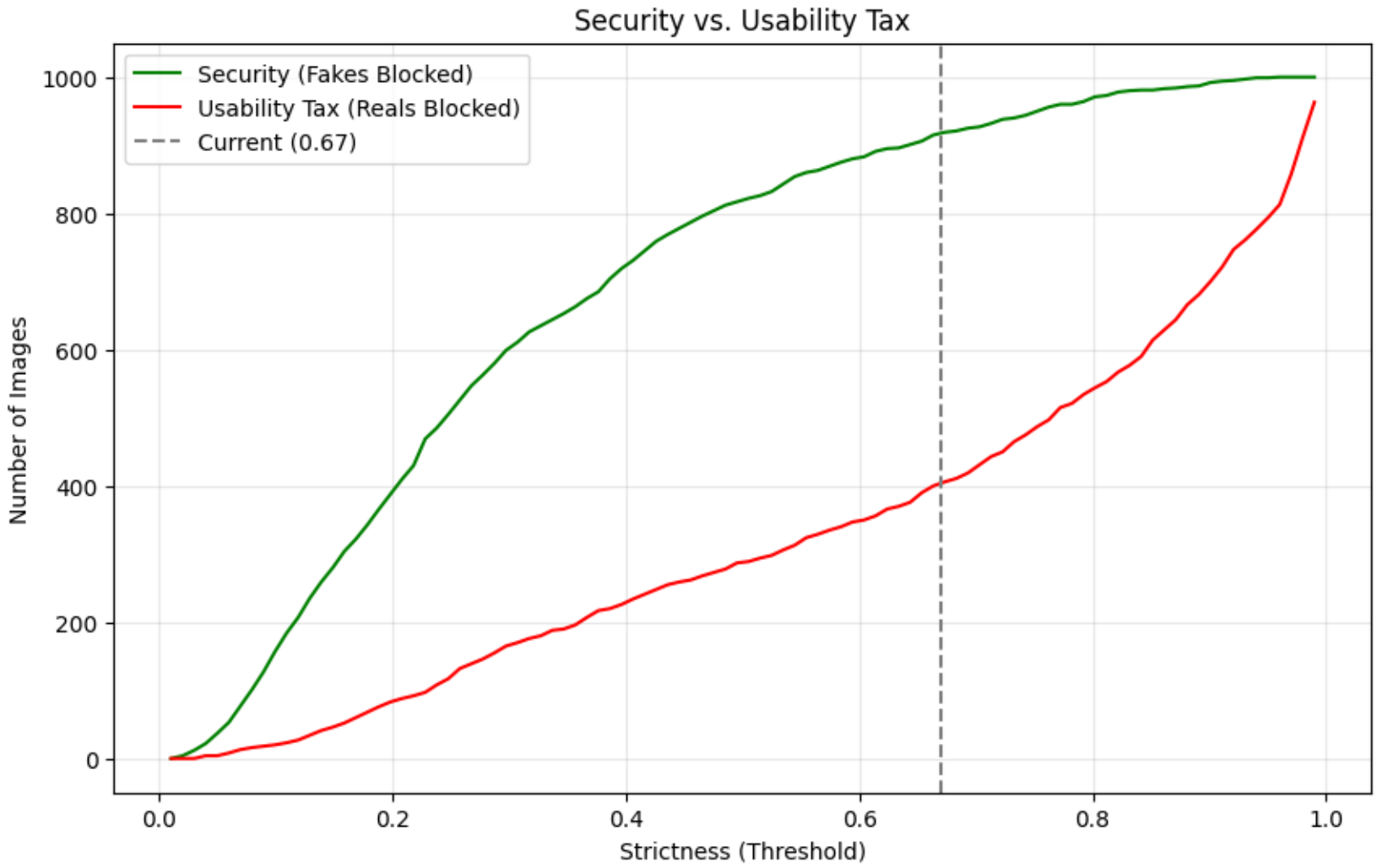
Item 4 - Feature Correlation 41, 141, 212



Item 5 - 141 Feature ROC

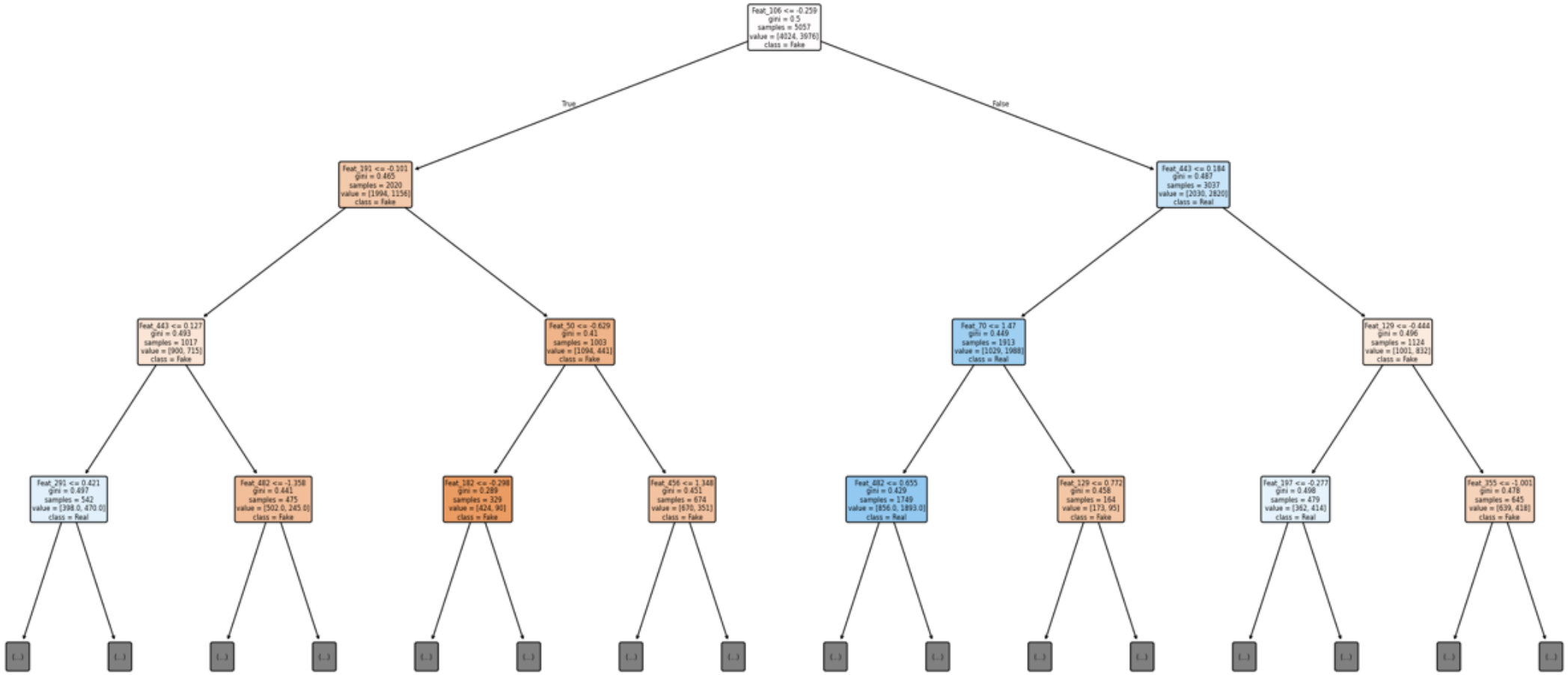


Item 6 - Security vs. Usability Tax, 141 SVM

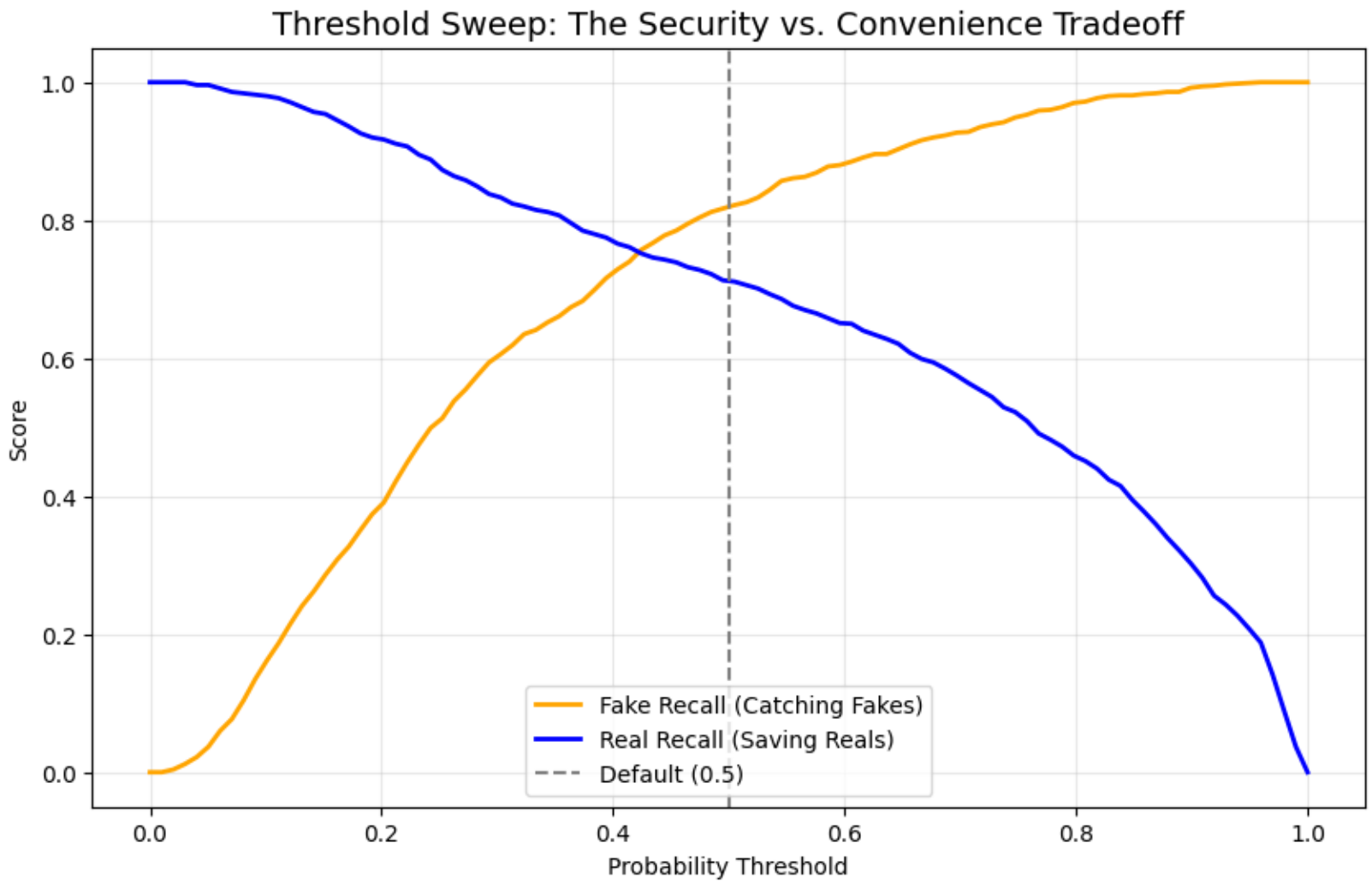


Item 7 - Feature Separation Logic

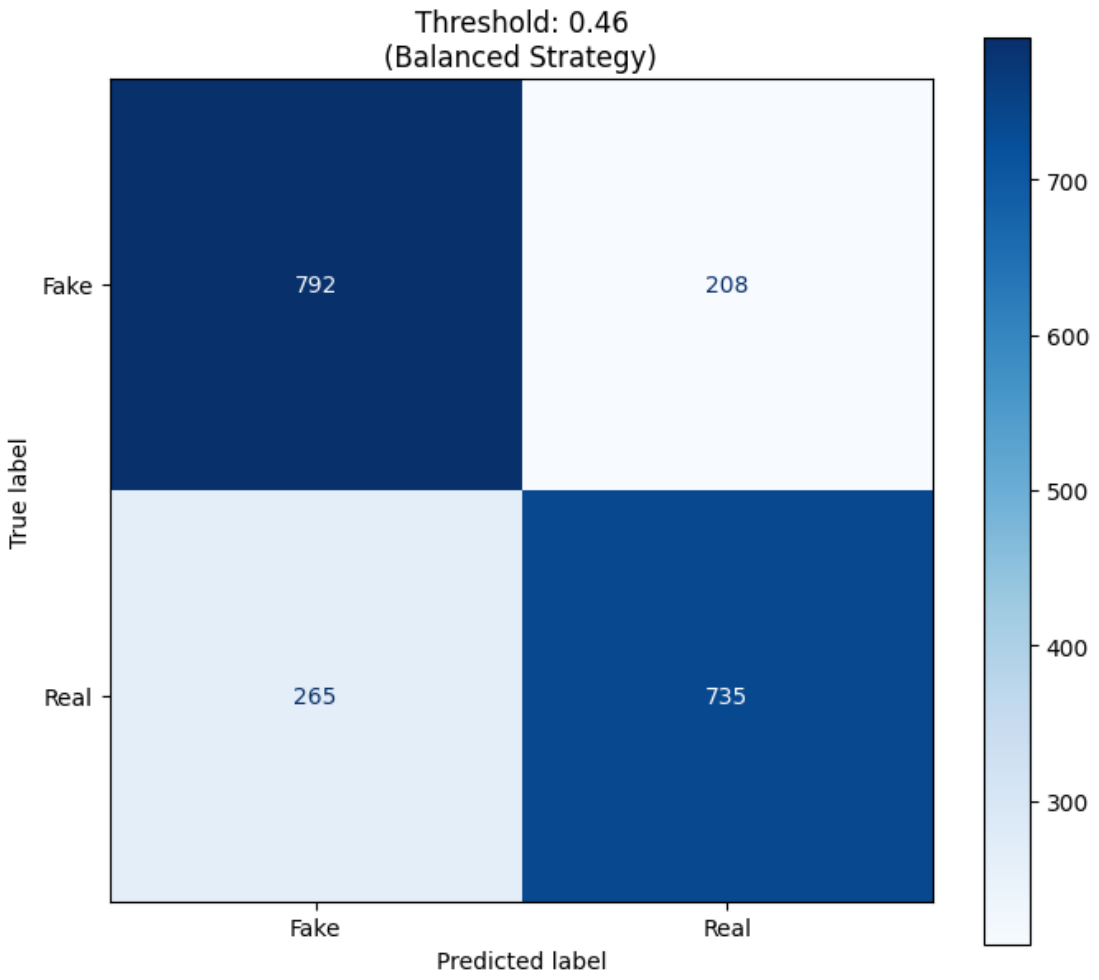
Decision Logic: How the Model Splits Real vs. Fake Faces (Top 3 Levels)

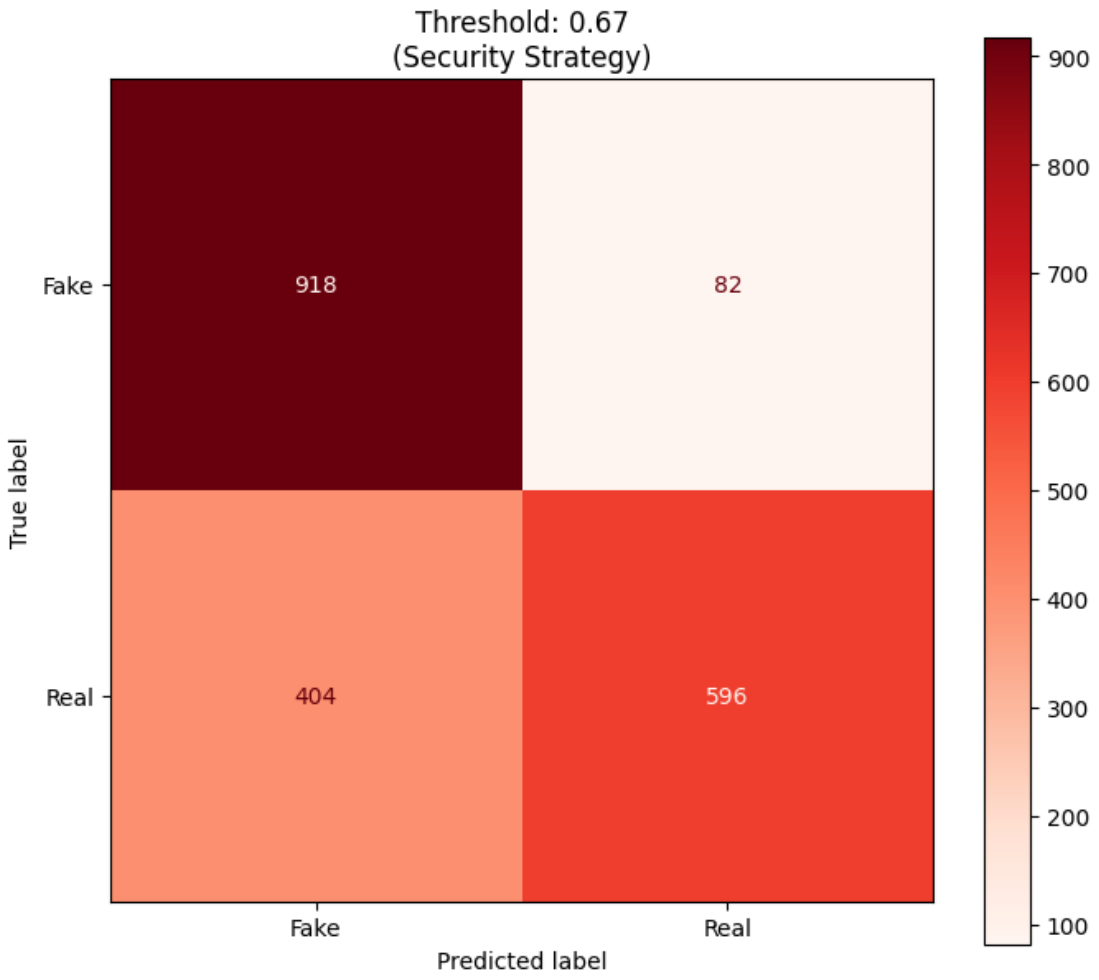


Item 8 - Threshold Sweep, 141 SVM

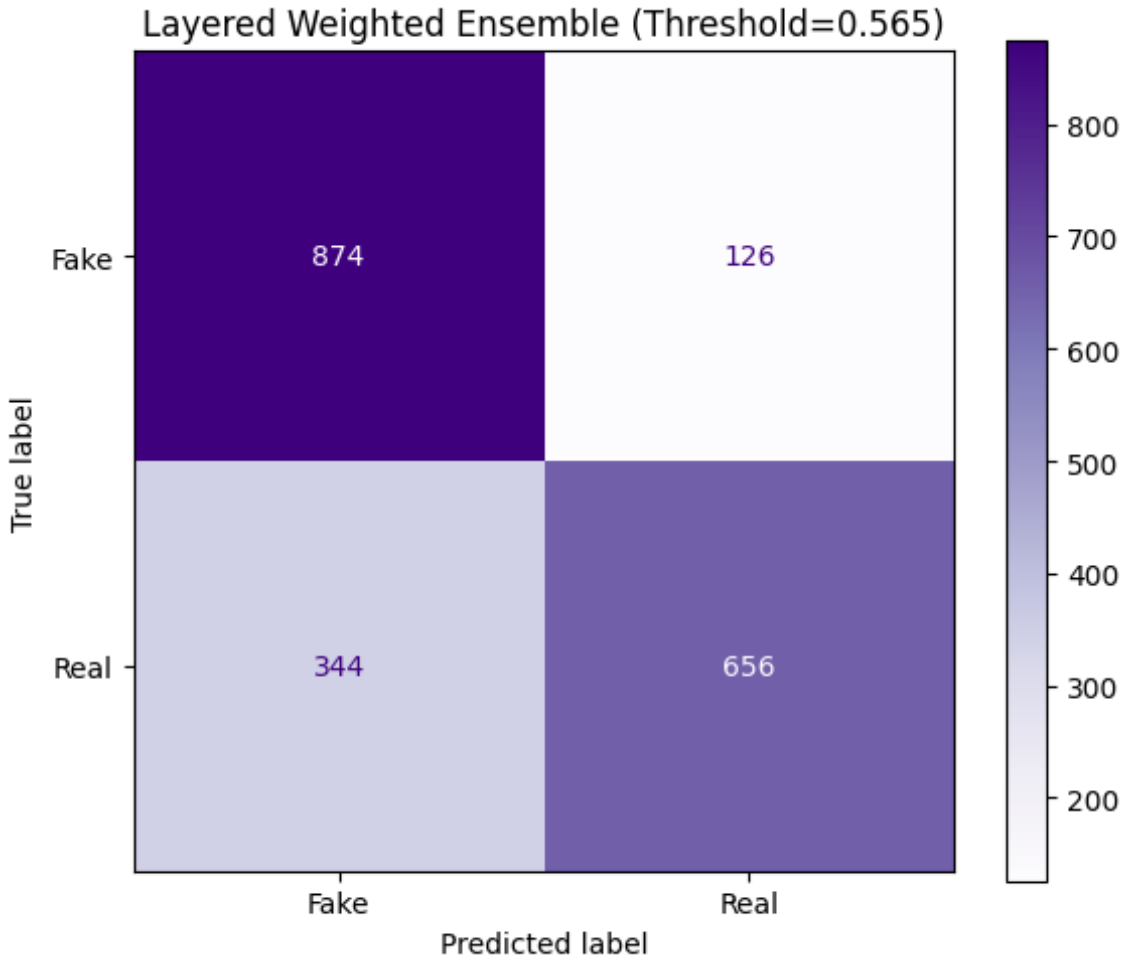


Item 9 - Balanced vs. Security Threshold

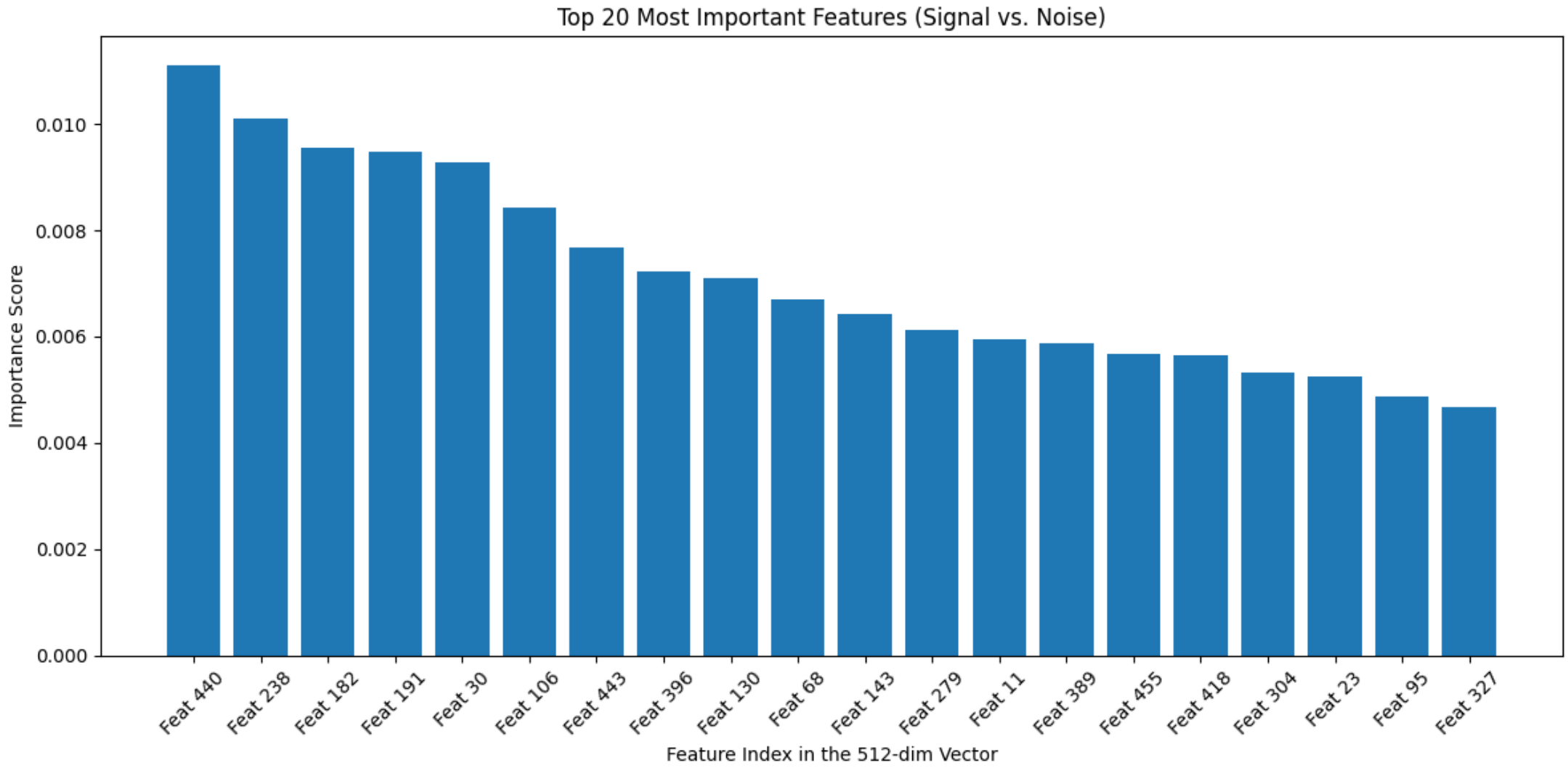




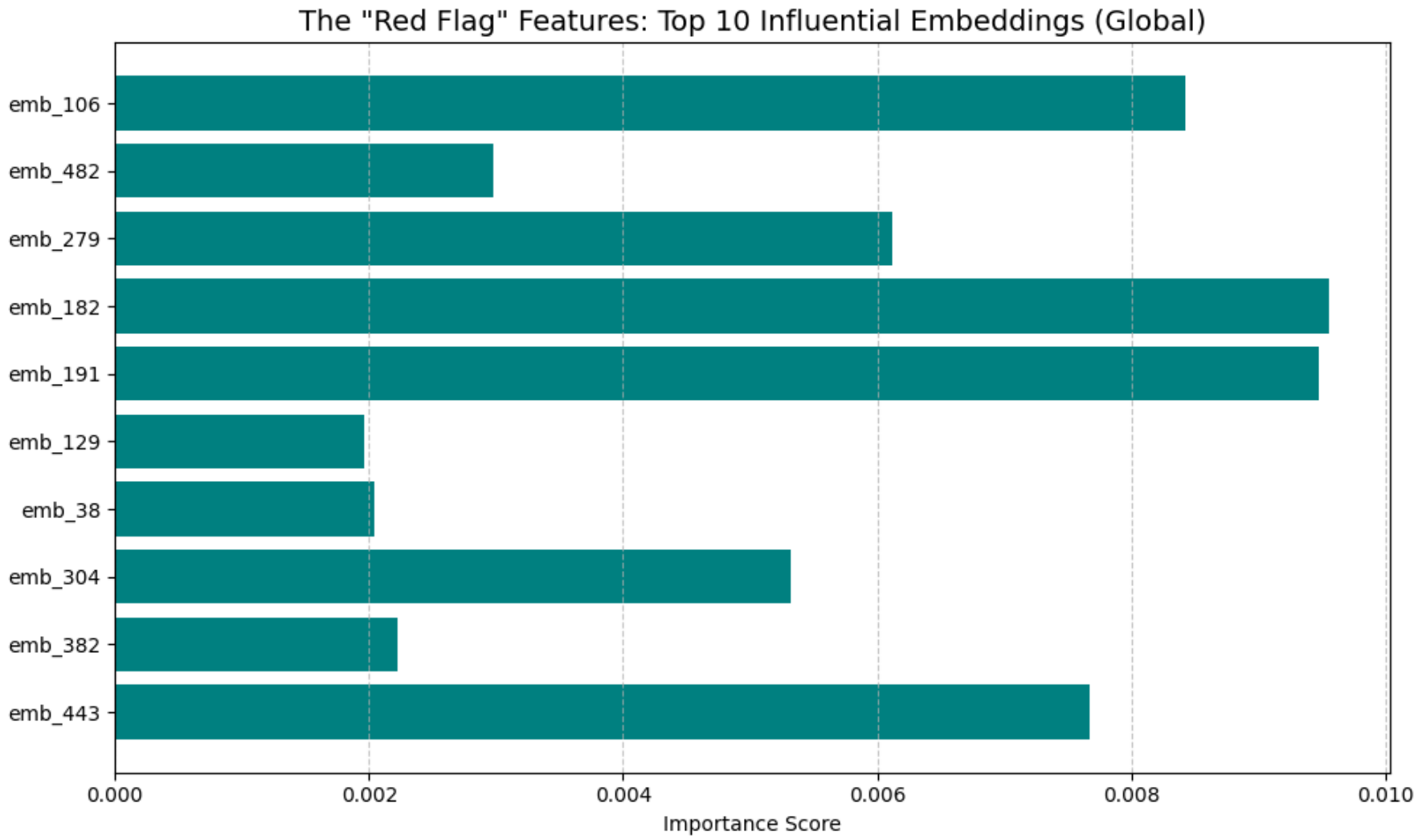
Item 10 - Layered Weighted Ensemble



Item 11 - Most Important Features



Item 12 - Global Red Flag Features



Item 13 – Full Model Comparison Table

Model	512 Features	200 Features	100 Features	20 Features
Support Vector Machine	0.78	0.77	0.76	0.74
MLP	0.76	0.75	0.75	0.74
Linear SVC	0.76	0.74	0.73	0.71
Logistic Regression	0.76	0.74	0.73	0.71
Stochastic Gradient Descent	0.76	0.71	0.66	0.67
CNN	0.75	0.71	0.73	0.70
Perceptron	0.74	0.68	0.67	0.63
Gradient Boosting	0.73	0.72	0.72	0.72
Random Forest	0.72	0.70	0.72	0.71
K-Nearest Neighbors	0.65	0.67	0.68	0.67
Naive Bayes	0.63	0.62	0.63	0.65
Decision Tree	0.62	0.62	0.61	0.61